

MONCTON ROMEO LEBLANC, NB, Canada

WMO: 717050

Lat: 46.1053N

Lon: 64.6839W

Elev: 71

StdP: 100.48

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.5	(c) -19.2	(d) -27.4	(e) 0.3	(f) -20.8	(g) -24.7	(h) 0.4	(i) -18.4	(j) 15.5	(k) -5.8	(l) 13.9	(m) -5.9	(n) 5.5	(o) 260	(p) 0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	28.8	21.0	27.2	19.9	25.7	19.1	22.3	26.8	21.4	25.3	20.4	24.0	5.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.6	24.4	20.0	14.8	23.6	19.1	14.0	22.5	65.9	26.8	62.5	25.4	59.1	24.0	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	11.1	9.8	-24.8	32.0	2.9	1.5	-26.9	33.1	-28.6	34.0	-30.3	34.8	-32.4	35.8
			-25.1	24.2	2.8	0.8	-27.2	24.7	-28.8	25.2	-30.4	25.6	-32.5	26.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-7.8	-6.9	-2.4	3.9	10.2	15.5	19.5	18.8	14.6	8.3	2.4	-3.8		
	DBStd	10.48	6.44	5.82	4.93	4.02	3.88	3.68	2.65	2.84	3.87	4.09	4.97	5.54		
	HDD10.0	2394	553	472	384	189	45	3	0	0	6	82	233	427		
	HDD18.3	4592	812	706	642	434	253	100	19	29	123	311	478	686		
	CDD10.0	969	0	0	1	5	52	167	293	273	143	31	5	0		
	CDD18.3	124	0	0	0	0	1	13	54	44	11	1	0	0		
	CDH23.3	977	0	0	1	2	33	139	408	318	75	2	0	0		
	CDH26.7	193	0	0	0	0	4	25	84	66	15	0	0	0		
(14) Wind	WSAvg	4.7	5.4	5.2	5.4	5.1	4.6	4.2	3.8	3.8	4.1	4.6	5.0	5.3		
(15) Precipitation	PrecAvg	1290	128	105	127	107	91	97	106	82	83	108	117	136		
	PrecMax	1612	284	166	201	239	223	172	216	180	158	218	231	265		
	PrecMin	840	19	41	27	36	23	24	19	14	15	42	34	46		
	PrecStd	191	59	29	42	51	47	39	51	38	36	46	43	53		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	10.1	13.4	20.2	26.9	29.4	30.9	30.8	28.6	22.9	17.7	12.8		
		MCWB	11.6	8.6	8.7	13.7	16.6	20.4	22.5	21.9	21.3	17.3	15.8	11.8		
	2%	DB	7.5	6.1	9.1	16.1	23.4	26.7	28.9	28.5	25.3	19.3	15.2	9.2		
		MCWB	6.7	4.8	6.4	11.4	14.9	18.9	21.4	20.7	19.4	15.2	13.7	8.5		
	5%	DB	4.0	3.5	6.7	13.3	20.5	24.8	27.3	26.7	23.2	17.2	13.0	6.1		
		MCWB	2.9	2.0	4.4	9.5	13.6	17.9	20.3	19.8	18.4	14.5	11.1	5.2		
	10%	DB	1.1	1.4	4.5	10.8	17.8	22.7	25.7	25.0	21.2	15.2	10.0	3.6		
		MCWB	0.3	0.4	2.4	7.1	12.4	16.7	19.4	19.0	17.4	12.5	8.0	2.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	8.9	10.4	15.4	18.8	22.2	23.7	23.4	22.2	18.5	16.5	11.8		
		MCDB	12.4	10.0	11.8	18.3	23.9	27.3	29.3	28.0	26.9	20.5	17.6	12.5		
	2%	WB	6.8	5.1	7.3	12.6	16.6	20.6	22.5	22.1	20.7	16.6	14.1	8.7		
		MCDB	7.3	5.9	8.6	14.8	20.9	24.7	27.3	26.2	23.7	18.3	14.9	9.3		
	5%	WB	3.1	2.3	4.8	10.0	14.9	19.2	21.5	21.1	19.5	14.9	11.4	5.4		
		MCDB	3.8	3.3	6.2	12.6	18.9	22.7	25.6	24.9	22.1	17.0	12.7	6.0		
	10%	WB	0.3	0.5	2.6	7.6	13.1	17.8	20.5	20.1	18.1	12.9	8.4	2.8		
		MCDB	0.9	1.3	4.3	10.5	16.7	21.3	24.1	23.5	20.6	14.6	9.7	3.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.1	8.5	9.0	10.8	10.7	10.5	10.6	10.5	9.2	7.5	7.6		
		MCDBR	13.2	10.3	11.0	13.6	16.1	14.4	13.2	13.2	12.6	12.0	10.5	10.1		
	5% WB	MCWBR	13.1	9.4	9.0	9.4	9.1	8.0	6.8	6.7	7.4	8.6	9.6	9.9		
		MCWBR	13.1	10.0	10.4	11.9	13.3	11.9	11.3	11.0	10.7	10.4	10.1	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.287	0.284	0.319	0.344	0.366	0.387	0.414	0.388	0.354	0.326	0.306	0.284		
	taud		2.413	2.399	2.402	2.419	2.403	2.343	2.302	2.370	2.452	2.530	2.530	2.496		
	Ebn at Noon		820	901	911	915	900	879	849	859	862	842	796	789		
	Edh at Noon		70	86	100	108	114	122	125	113	95	76	63	59		
(44) All-Sky Solar Radiation	RadAvg		1.43	2.19	3.38	4.24	4.97	5.37	5.52	4.86	3.88	2.45	1.37	1.06		
	RadStd		0.13	0.20	0.22	0.41	0.52	0.43	0.37	0.33	0.32	0.25	0.16	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	+0.38	+0.47	N/A	N/A	+65	+24		
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.34	N/A	N/A	N/A	N/A		

Nomenclature: See separate page